

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: **Recommend Endpoint Security Measures**

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Objectives

- **Part 1:** Recommend Mitigation Procedures to Address Vulnerabilities
- **Part 2:** Recommend an Endpoint Protection Product for a New Network

Background / Scenario

To provide security, professionals typically implement a number of network security measures that work together in a layered security approach. Firewalls and other devices protect the network perimeter from attack; however, it is always possible that threats can elude these defenses. Therefore, it is not only necessary to protect the network perimeter, but also take action to protect individual network hosts from compromise. In this lab, two case studies are examined and endpoint threat mitigations are recommended to address the exploits identified.

Part 1: Recommend Mitigation Procedures to Address Vulnerabilities

A vulnerability assessment of a manufacturing company's network was completed and the following five endpoint security issues were identified. The table below documents each vulnerability alongside its associated risks and the recommended mitigation procedures.

Vulnerability	Risk / Exposure	Recommended Mitigation Procedures
SCADA Systems Running on Windows XP	Windows XP has been end-of-life since April 2014 — no security patches are released. SCADA systems on this OS are exposed to all known Windows XP vulnerabilities with no remedy.	• Upgrade the SCADA operating system to a supported version of Windows (Windows 10/11 LTSC recommended for industrial systems). • If upgrading is not immediately possible, isolate SCADA systems on a separate, air-gapped or heavily segmented network VLAN. • Deploy application

		whitelisting to allow only approved SCADA software to execute. • Apply a host-based firewall with strict inbound/outbound rules to limit exposure. • Engage the SCADA vendor for a supported OS migration roadmap and any available security patches for legacy systems. • Implement network monitoring and anomaly detection around the SCADA segment.
Use of Unknown USB Media on Critical Systems	Unknown USB drives can introduce malware (e.g. Stuxnet targeted SCADA systems via USB). Uncontrolled removable media is a leading vector for data theft and ransomware deployment.	• Implement a removable media policy that prohibits the use of unauthorised USB drives on all critical systems. • Use Group Policy (GPO) or endpoint management tools to disable USB ports on critical systems where removable media is not operationally required. • Where USB use is necessary, deploy a Data Loss Prevention (DLP) solution to whitelist only approved, encrypted devices. • Enable logging and auditing of all USB connection events. • Conduct regular user awareness training on the risks of unknown removable media.

Personal Devices Accessing the Network (BYOD)	Personal smartphones, tablets, and laptops may not meet the organisation's security standards. They may lack patches, carry malware, and introduce uncontrolled access points to the network.	• Implement a Network Admission Control (NAC) solution to assess device compliance before granting network access. • Establish and enforce a formal BYOD policy defining minimum security requirements (OS version, antivirus, screen lock, encryption). • Segment BYOD devices onto a dedicated guest/BYOD VLAN, isolated from critical systems and servers. • Require all personal devices to be enrolled in a Mobile Device Management (MDM) solution. • Use multi-factor authentication (MFA) for all network access from personal devices. • Regularly audit connected devices and revoke access for non-compliant or unrecognised endpoints.
Unrestricted Web Browsing Including Malware Sites	Free access to malicious websites exposes users to drive-by downloads, phishing pages, and exploit kits. This is a primary delivery mechanism for malware and ransomware.	• Deploy a Web Security Appliance (WSA) or web proxy to filter and monitor all outbound web traffic. • Configure DNS-based filtering (e.g. Cisco Umbrella) to block access to known malicious domains at the DNS layer. • Implement and enforce an

		Acceptable Use Policy (AUP) that prohibits access to inappropriate or high-risk websites. • Enable safe browsing features within the host-based security suite on all endpoints. • Configure the web filter with category-based blocking for high-risk site types (malware, phishing, adult content, anonymisers). • Provide regular security awareness training to help users identify phishing and suspicious websites.
Inconsistent / Legacy Antivirus with Unknown Signature Status	Mixed and outdated antivirus software with stale signatures leaves hosts unable to detect recent malware. Inconsistency also creates management blind spots across the environment.	• Standardise on a single, enterprise-grade antimalware solution across all hosts (e.g. Cisco AMP, CrowdStrike, or Microsoft Defender for Endpoint). • Ensure all antivirus agents are configured for automatic signature updates — ideally multiple times per day. • Deploy a centralised endpoint management console to monitor protection status, update compliance, and run scans remotely. • Conduct an immediate full scan of all endpoints to identify and remediate any existing infections. • Remove all legacy or

		unsupported antivirus products and replace with the standardised solution. • Implement heuristics-based and behaviour-based detection alongside signature-based scanning for protection against zero-day threats.
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Part 2: Recommend an Endpoint Protection Product for a New Network

A friend has recently received venture capital funding for a promising new product. Rapid growth is predicted, and a new office location is being opened. The following recommendation provides a comprehensive endpoint security product suited to a small but fast-growing startup environment.

Recommended Product: **CrowdStrike Falcon (Go / Pro tier)**

Justification and Reasoning

CrowdStrike Falcon is a cloud-native endpoint detection and response (EDR) platform that is well-suited to a startup environment. It is lightweight, quick to deploy, scalable, and does not require dedicated on-premise hardware — making it ideal for a small team that will grow.

Key Features Supporting This Recommendation

- **Cloud-Native Architecture:** Falcon is entirely cloud-delivered. There is no on-premise server to set up or maintain. This is critical for a startup — the security team (or even a single person) can manage all endpoints from a web console without needing dedicated infrastructure.
- **Lightweight Agent:** The Falcon sensor is a single, lightweight agent installed on each endpoint. It has minimal impact on system performance — employees will not notice it running. This matters for a startup where productivity is paramount.

- **AI and Behaviour-Based Detection:** Falcon uses machine learning and behaviour-based analysis to detect threats — including zero-day and fileless malware that signature-based tools miss entirely. This gives strong protection even before new threat signatures are published.
- **Scalability:** As the company grows from a handful of employees to potentially hundreds, Falcon scales seamlessly. Adding a new endpoint simply requires installing the agent — no reconfiguration of the central platform needed.
- **Threat Intelligence Integration:** CrowdStrike Intelligence (CrowdStrike's threat research arm) continuously feeds updated threat intelligence into the Falcon platform. Endpoints are always protected against the latest known threats worldwide.
- **EDR Capability:** Beyond prevention, Falcon provides full endpoint detection and response (EDR). If an incident occurs, security personnel can investigate exactly what happened — which processes ran, which files were accessed, which network connections were made — and contain the threat.
- **Centralised Management Console:** A single web-based dashboard provides visibility across all endpoints. Alerts, detections, and device health are all visible in one place — no need to check each machine individually.
- **Compliance Support:** As the startup grows and potentially enters regulated industries, Falcon supports compliance requirements (PCI-DSS, HIPAA, SOC 2) through its reporting and logging capabilities.

Consideration for Growth

CrowdStrike Falcon Go is an affordable entry point for small teams, covering antivirus and malware protection with the same cloud infrastructure. As the company grows, upgrading to Falcon Pro or Falcon Enterprise adds full EDR, threat hunting, and identity protection without changing the platform or re-deploying agents. This protects the initial investment and avoids disruptive technology migrations during periods of rapid growth.

Alternative Considered

Microsoft Defender for Endpoint was also considered, particularly given its integration with Microsoft 365 which many startups already use. However, CrowdStrike was selected due to its stronger independent detection rates in third-party testing (AV-TEST, SE Labs), its superior EDR capabilities at the entry-level tier, and its platform-agnostic support for Windows, macOS, and Linux — important for a modern startup environment where developers may use diverse operating systems.

Conclusion

This lab demonstrated that endpoint security requires a multi-layered approach. A single product or policy is never sufficient — effective protection combines operating system hygiene (patching, supported OS versions), access controls (USB policies, NAC, BYOD policies), web filtering (WSA, DNS filtering), and comprehensive endpoint security software (antimalware with behaviour-based detection, host-based firewalls, EDR).

For the manufacturing company in Part 1, the identified vulnerabilities represent significant and immediate risk — particularly the unsupported SCADA operating system and unknown USB media usage. These must be addressed as a priority before the network can be considered compliant with stringent security standards.

For the startup in Part 2, CrowdStrike Falcon provides a scalable, cloud-native endpoint security platform that grows with the business — offering strong protection from day one without requiring dedicated security infrastructure or personnel.